

## Task-6

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP$ cd "2520030103_Practical"
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical$ mkdir linux_task4
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical$ cd linux_task4
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ touch public.txt private.txt script.sh
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ echo "This is public file" > public.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ echo "This is private file" > private.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ echo "This is a shell script" > script.sh
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ ls -l
total 12
-rw-r--r-- 1 poojitha_sai poojitha_sai 21 Aug 12 15:41 private.txt
-rw-r--r-- 1 poojitha_sai poojitha_sai 20 Aug 12 15:41 public.txt
-rw-r--r-- 1 poojitha_sai poojitha_sai 23 Aug 12 15:41 script.sh
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ chmod 600 private.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ chmod 644 public.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ chmod +x script.sh
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ ls -l
total 12
-rw----- 1 poojitha_sai poojitha_sai 21 Aug 12 15:41 private.txt
-rw-r--r-- 1 poojitha_sai poojitha_sai 20 Aug 12 15:41 public.txt
-rwxr-xr-x 1 poojitha_sai poojitha_sai 23 Aug 12 15:41 script.sh
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ chmod -x script.sh
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ ls -l
total 12
-rw----- 1 poojitha_sai poojitha_sai 21 Aug 12 15:41 private.txt
-rw-r--r-- 1 poojitha_sai poojitha_sai 20 Aug 12 15:41 public.txt
-rw-r--r-- 1 poojitha_sai poojitha_sai 23 Aug 12 15:41 script.sh
```

## Task-7

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_task4$ cd ..
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical$ mkdir permission_lab
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical$ cd permission_lab
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ touch test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ echo "This is test content" > test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ chmod -r test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ cat test.txt
cat: test.txt: Permission denied
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ cat test.txt
cat: test.txt: Permission denied
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ |
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ chmod -r test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ cat test.txt
cat: test.txt: Permission denied
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ |
```

## Task - 8

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ chmod +r test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ cat test.txt
Hello, this is a permission test.
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ |
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ cat test.txt
Hello, this is a permission test.
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ chmod -w test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ echo "New content" >> test.txt
-bash: test.txt: Permission denied
```

```
cat: test.txt: Permission denied
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ chmod +r test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ cat test.txt
Hello, this is a permission test.
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ chmod -w test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ echo "New content" >> test.txt
-bash: test.txt: Permission denied
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ chmod +w test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ echo "New content" >> test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ chmod -x test.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/permission_lab$ |
```

## Task-9

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical$ mkdir linux_project
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical$ |
```

```
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical$ cd linux_project
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ mkdir source backup docs
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ touch source/main.c source/helper.c source/README.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ ls
backup docs source
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ |
```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ cat > source/README.txt <<EOF
Line 1: Linux Project
Line 2: This is a sample project
Line 3: Learning Linux commands
Line 4: Working with directories
Line 5: Working with files
Line 6: Creating files
Line 7: Copying files
Line 8: Renaming files
Line 9: Changing permissions
Line 10: Reading file contents
Line 11: Using head command
Line 12: Using tail command
Line 13: Using mkdir command
Line 14: Using touch command
Line 15: Using cp command
Line 16: Using mv command
Line 17: Using chmod command
Line 18: Using rm command
Line 19: Using rmdir command
Line 20: Linux commands practice
EOF
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ |

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ head -n 5 source/README.txt
Line 1: Linux Project
Line 2: This is a sample project
Line 3: Learning Linux commands
Line 4: Working with directories
Line 5: Working with files
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ tail -n 5 source/README.txt
Line 16: Using mv command
Line 17: Using chmod command
Line 18: Using rm command
Line 19: Using rmdir command
Line 20: Linux commands practice
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ |

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ cp source/main.c backup/
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ cp source/README.txt docs/
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ |

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ mv source/headers.c source/functions.c
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ chmod 644 source/README.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ ls
backup docs source
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ |

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ rm backup/main.c
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ rmdir backup
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ |

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ rmdir backup
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$ ls -R
.:
docs source

./docs:
README.txt

./source:
README.txt functions.c main.c
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/linux_project$

```

## Task-10

```

└─ subsystemcalls

26 directories, 47 files
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical$ mkdir student_portal
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical$ cd student_portal
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ |

```

```

onfig
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch src/main.c
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch src/student.c
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch docs/README.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch config/config.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch logs/system.log
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch src/main.c
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch src/student.c
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch docs/README.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch config/config.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ touch logs/system.log
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ cat > logs/system.log << EOF

```

```

Student Portal System Started
Checking configuration
Loading student database
Database connection successful
Loading student records
Student authentication enabled
Admin authentication enabled
Checking system permissions
Backup service initialized
Log service initialized
Student module loaded
Course module loaded
Attendance module loaded
Marks module loaded
Student Portal System Ready
EOF

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ |

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ head -n 5 logs/system.log
Student Portal System Started
Checking configuration
Loading student database
Database connection successful
Loading student records

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ tail -n 5 logs/system.log
Student module loaded
Course module loaded
Attendance module loaded
Marks module loaded
Student Portal System Ready
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ |

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ cp docs/README.txt backup/
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ |

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ cp docs/README.txt backup/
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ mv backup/README.txt backup/
README_backup.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ |

```

```

README_backup.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ chmod 600 config/config.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ chmod 644 docs/README.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ |

```

```

README_backup.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ chmod 600 config/config.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ chmod 644 docs/README.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ rm backup/README_backup.txt
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ rmdir backup
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ |

```

```

poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ ls -R
.:
config docs logs src

./config:
config.txt

./docs:
README.txt

./logs:
system.log

./src:
main.c student.c
poojitha_sai@LAPTOP-8L1DST90:~/Thread Runners_OSSP/2520030103_Practical/student_portal$ |

```